Describe a time when you had a positive impact on others. What were the challenges? What were the rewards? ( 500-650)

“What are you reading?”

“Microcontroller from the ICT book.”

“Why are you reading that? Didn’t your class teacher demonstrate it in the lab with a microcontroller?”

“They don’t even take the ICT class seriously, let alone the lab.”

This thought-provoking conversation with my little sister made me realise the significant gap in the Bangladeshi STEM Education. Driven to make a difference in STEM Sector, I founded FTP (Future-Talent-Programme) bootcamp, where I organised an annual three-month-long program to teach students C++ and Python with hands-on DIY projects with microcontrollers like Arduino and NodeMcu. I also mentored students for national and global hackathons to help them gain practical experience by solving real-world challenges. The team I mentored this year for **NASA-International-Space-Apps-Challenge** made it the global stage as a Global Nominee.

Little did I know that the bootcamp I started to share knowledge would end up teaching me lifelong lessons and invaluable experiences.

It’s a new batch at FTP'23, I was teaching just like I had in previous batches.

“...you use the Wire.beginTransmission() function to send commands to this PCA9685's address.”
“Did you understand?” I asked, and Nabila nodded along with everyone else but her expression betrayed her.

Many students like her in that session were hesitant to ask questions due to embarrassment. This made me realize my lack of effective communication and the need to adjust my teaching approach. That’s when I came up with the "letterbox" concept, which allowed students to leave anonymous notes while maintaining their privacy. In the upcoming class, I revised the topics mentioned in the notes. This method worked like a charm, as I started receiving a continuous stream of notes from then on.

I have taught around 150 students over the past two years, many of whom were the same age as my little sister while encountering countless challenges along the way. It wouldn’t be wrong to say I have acquired around 150 skills, as every student requires a unique approach to learning and engagement based on their individual needs and styles. For instance, Nabila taught me to be more patient and adapt my teaching methods to fit different learning paces, while Dipto taught me the importance of positive reinforcement, to name a few.

Teaching at FTP for the last two years has allowed me to lead with empathy, adapt to challenges, and acquire diverse teaching skills to address every student's needs while recognizing my shortcomings.

I even developed my own philosophy: mentoring outlives the mentor. The students I mentor today will, in turn, mentor others in the future, creating a long-lasting chain of knowledge. In the future, whatever community I become part of, I want to keep engaged in this type of mentoring.

In the future, whatever community I become part of, I want to take a similar role by volunteering at outreach programs. Given my diverse experience with kids and high school students from FTP and navigating the differences and challenges, I am sure I will bring new perspectives to those initiatives.

[Amherst College]()

****Important Univeristy For Aid***

Please briefly elaborate on an extracurricular activity or work experience of particular significance to you. (Maximum: 175 words)*

“Damn, Sakib! You said it... and it happened! Amazing boy!” That was one of the compliments I received after becoming a Global Finalist at the NASA Space Apps Challenge which is the world's largest hackathon. This achievement even made national headlines in my country: “Horizon BD Represents Bangladesh at NASA International Space Apps Challenge!”

Network outages are very common in Bangladesh during disasters, and as a result, crucial aid and medical supplies often can't reach affected areas. I tried to address this issue in the hackathon through my hardware and software solution, SDRAM (Sustainable-Disaster-Response-Alert-Mechanism) where I have used a LoRa SX1278 module and NodeMCU connected to my React Native app via a Wi-Fi access point. This system provided communication over 12 km through a half-duplex method. The app also featured a trained sequential neural network to predict lightning and provide alerts for disasters like floods, cyclones, and typhoons.

Having my work appreciated on such a global stage, and knowing that it might help countless people someday, makes this achievement especially meaningful to me.

What do you see as the benefits of linking learning with leadership and/or service? In your response, please share with us a time where you have seen that benefit through your own experience.(350)

#FTP

A thought-provoking conversation with my little sister made me realise the significant gap in the Bangladeshi STEM Education. Driven to make a difference in STEM Sector, I founded FTP (Future-Talent-Programme) bootcamp, where I organised an annual three-month-long program to teach students C++ and Python with hands-on DIY projects with microcontrollers like Arduino and NodeMcu. I also mentored students for national and global hackathons to help them gain practical experience by solving real-world challenges. The team I mentored this year for **NASA-International-Space-Apps-Challenge** made it the global stage as a Global Nominee.

Little did I know that the bootcamp I started to share knowledge would end up teaching me lifelong lessons and invaluable experiences.

It’s a new batch at FTP'23, I was teaching just like I had in previous batches.

“...you use the Wire.beginTransmission() function to send commands to this PCA9685's address.”
“Did you understand?” I asked, and Nabila nodded along with everyone else but her expression betrayed her.

Many students like her in that session were hesitant to ask questions due to embarrassment. This made me realize my lack of effective communication and the need to adjust my teaching approach. That’s when I came up with the "letterbox" concept, which allowed students to leave anonymous notes while maintaining their privacy. In the upcoming class, I revised the topics mentioned in the notes. This method worked like a charm, as I started receiving a continuous stream of notes from then on.

I have taught around 150 students over the past two years, many of whom were the same age as my little sister while encountering countless challenges along the way. It wouldn't be wrong to say I have acquired around 150 skills, as every student requires a unique approach to learning and engagement based on their individual needs and styles. For instance, Nabila taught me to be more patient and adapt my teaching methods to fit different learning paces, while Dipto taught me the importance of positive reinforcement, to name a few.

Teaching at FTP has not only allowed me to lead but has also shown me how to lead with empathy and adaptability while acquiring diverse teaching skills.

#Extra:

I started my computing journey with MIT-App-Inventor at 13. Later I self-taught Java to build Android Applications. My curiosity grew with my age and led me to engage in other fields of CS through projects and hackathons.

My LoRa (SX1278-432 MHz) module-based communication and early-disaster-prediction-system helped me crack NASA-International-Space-Apps-Challenge which solved communication issues in Bangladesh during the disaster. In my BCI project I captured neural signals with a single-channel amplifier and Raspberry-Pi-Zero. Recently, I built an autonomous drone with a Raspberry-Pi-Zero navigated by a multimodal model.

In FTP-Bootcamp I tried to share this passion through hands-on DIY projects with others.

While engaging on those hackathons and projects made me realize the importance of collaboration and mentorship.

Amherst's Summer Science Undergraduate Research Fellowship (SURF) Program will provide me with opportunities to **collaborate** and get mentor support in research across various fields, including neuroscience, robotics, data science, computer vision, embedded systems, natural language processing, and cyber-physical systems. For someone passionate like me, these opportunities are like a playground of endless possibilities.

#Research:

The lack of accessible early warning systems for lightning led me to develop a DNN model using environmental parameters from NASA's open data archive, including Temp(2M), Dew(2M), Pressure(2M), Relative Humidity(2M) and others.

I faced imbalanced datasets accross the label due to rare lightning events.

To address the imbalance, I used Synthetic-Minority-Over-sampling-Technique (SMOTE).

The model achieved an F1 score of 92% for both classes: 0 (not likely to strike) and 1 (likely to strike).

## Bowdoin College

******Important Univeristy For Aid*****

If you wish, you may share anything about the unique experiences and perspectives that you would bring with you to the Bowdoin campus and community or an experience you have had that required you to navigate across or through difference. (250 words)

#FTP

A thought-provoking conversation with my little sister made me realise the significant gap in the Bangladeshi STEM Education. Driven to make a difference in STEM Sector, I founded FTP (Future-Talent-Programme) bootcamp, where I organised an annual three-month-long program to teach students C++ and Python with hands-on DIY projects with microcontrollers like Arduino and NodeMcu. I also mentored students for national and global hackathons to help them gain practical experience by solving real-world challenges. The team I mentored this year for **NASA-International-Space-Apps-Challenge** made it the global stage as a Global Nominee.

I have taught around 150 students oer the pastv two years, many of whom were the same age as my little sister while encountering countless challenges along the way. It wouldn't be wrong to say I have acquired around 150 skills, as every student requires a unique approach to learning and engagement based on their individual needs and styles. For instance, Nabila taught me to be more patient and adapt my teaching methods to fit different learning paces, while Dipto taught me the importance of positive reinforcement, to name a few.

At Bowdoin, I want to apply these skills through initiatives like the **Winter Break Community Engagement Program (WBCEP)**, where I can teach at a K-12 school and engage in hands-on projects with adaptive learning. Also I aspire to serve as a peer tutor at the **Baldwin Center for Learning and Teaching (BCLT)**, contributing to **Wicked Smart Groups and Workshops**.

**Which line from The Offer resonates most with you?***

To be at home in all lands and all ages

June 14, 2024, 2:35 PM. It's my 17th day in **Meghalaya**, the Land of Clouds, it was raining—just as the name suggests. My car moved upward, winding through the hills and mountains, making its way into the clouds. To have lunch I settled on a roadside dhaba. As I entered a Khashi waitress with a smile welcomed me with Khublei meaning "May God bless you". She was looking elegant in her tribal attire. I am quite fluent in Hindi, so to test my fluency, I asked for the menu in Hindi instead of English. "Menu mein kya hai?" I said. She smiled, perhaps amused to hear Hindi from a foreigner's mouth. She handed over the menu, and after glancing through it, I settled on **Minil Songa**, a unique traditional sticky rice dish steamed inside bamboo tubes. Before my food was served, I glanced through the window at the shops. Almost every shop here seemed to be run by women, reflecting its matrilineal society—a new experience for me. In a while, my food arrived.

I spent more than a month in **Meghalaya** before my departure on July 17, 2024. Looking back, the words **"To be at home in all lands and all ages"** resonate deeply with me, as I have encountered different tribal groups, such as the Khasi, Garo, and Jaintia, and seen the remarkable linguistic and cultural diversity. This line defines my journey of embracing diverse cultures, bridging gaps between traditions, and finding a sense of belonging in unfamiliar territories.

## Columbia University

List a selection of texts, resources and outlets that have contributed to your intellectual development outside of academic courses, including but not limited to books, journals, websites, podcasts, essays, plays, presentations, videos, museums and other content that you enjoy. (100 words or fewer)*

**Books:**
*Sapiens* by Yuval Noah Harari (favorite), *Zero to One* by Peter Thiel, *Deep Learning from Scratch* by Seth Weidman, *Guns, Germs, and Steel* by Jared Diamond, *The Three-Body Problem* by Liu Cixin

**Anime:**
*Code Geass*, *Violet Evergarden*, *Your Lie in April*, *Overlord*, *A Silent Voice*

**Websites:**
Kaggle, arXiv.org, GitHub, Medium, ResearchGate, Google Colab

**Talks and Conferences:**
TED Talks, IBM TechXchange Pre-Conference, Google I/O,

**Places:**
Bangabandhu Military Museum, Somapura Mahavihara, Mahasthangarh, Mainamati, Konark Sun Temple, Tawang Monastery

**Hackathons:**
NASA International Space Apps Challenge, Microsoft Imagine Cup, HackMIT

**Bootcamp/Community Services:**
FTP Bootcamp, where I teach students programming.

A hallmark of the Columbia experience is being able to learn and thrive in an equitable and inclusive community with a wide range of perspectives. Tell us about an aspect of your own perspective, viewpoint or lived experience that is important to you, and describe how it has shaped the way you would learn from and contribute to Columbia's diverse and collaborative community. (150 words or fewer)*

#FTP

A thought-provoking conversation with my little sister made me realise the gap in the Bangladeshi STEM-Education. Driven to make a difference in STEM Sector, I founded FTP (Future-Talent-Programme) bootcamp, where I organised an annual three-month-long program to teach students C++ and Python with hands-on DIY projects with microcontrollers like Arduino and NodeMcu. I also mentored students for national and global hackathons to help them gain practical experience by solving real-world-challenges. The team I mentored this year for **NASA-International-Space-Apps-Challenge** made it the global stage as a Global Nominee.

Little did I know that the bootcamp I started to share knowledge would end up teaching me lifelong lessons and invaluable experiences. I've learned to lead with empathy and adaptability while gaining diverse teaching skills.

At Columbia, I want to follow the same path by volunteering at initiatives like Triple F Empowerment and the Outreach Program by engaging K-12 students in STEAM activities.

In college/university, students are often challenged in ways that they could not predict or anticipate. It is important to us, therefore, to understand an applicant's ability to navigate through adversity. Please describe a barrier or obstacle you have faced and discuss the personal qualities, skills or insights you have developed as a result. (150 words or fewer)*

#FTP

"....use Wire.beginTransmission() function to send commands to this PCA9685's address."
"Did you understand?" Nabila nodded along with everyone else but her expression betrayed her confusion.

Many students like her in the bootcamp were hesitant to ask questions. I realized my lack of communication and the need to adjust my approach.
That's when I introduced the "letterbox" concept where students can leave anonymous notes. I revised the topics in the next class. This method worked as I began receiving a lot of notes.
So far I've taught 150 students each requiring a unique approach to learning while gaining invaluable skills and qualities.
For instance, Nabila taught me to be more patient and adapt to different learning paces, while Dipto showed me the importance of positive reinforcement.
Teaching at FTP for the last two years has allowed me to lead with empathy, adapt to challenges, and acquire skills while recognizing my shortcomings.

## Why are you interested in attending Columbia University? We encourage you to consider the aspect(s) that you find unique and compelling about Columbia. (150 words or fewer)*

While engaging in hackathon and project over the year I have realized that those projects could reach even higher milestones with collaboration and mentorship.

For example, my BCI project is actually inspired by the concept of Digital-Immortality which proposes that a person can achieve immortality by mimicking human consciousness in a virtual-world by mapping synaptic network. However, to achieve such a feat, I require extensive collaboration and mentorship across multiple disciplines.

This is where initiatives like the SURE and Collaboratory come in, where I can engage with other nerds like me and get mentorship on BCI projects as well as other innovative projects. Given my interest in AI and Neuroscience, I am eager to work in NeuroAI under the mentorship of Richard Zemel from Columbia University. It is incredibly exciting to explore how brain-inspired AI can enhance computational models and deepen our understanding of human cognition.

## What attracts you to your preferred areas of study at Columbia Engineering? (150 words or fewer)*

I started my computing journey with MIT-App-Inventor at 13. Later I self-taught Java to build Android Applications.

My curiosity and passion grew with my age which leads me to explore various fields of CS through projects and hackathons to solve real-world problems.

My LoRa (SX1278-432 MHz) module-based communication and early-disaster-prediction-system helped me to crack Nasa-Space-Apps which was intended to solve communication issues in Bangladesh during the disaster. My BCI project involved creating a single-channel amplifier with a NodeMCU to capture neural signals. Recently, I built an autonomous drone with an Raspberry-Pi-Zero navigated by a multimodal model hosted on Azure.

In FTP-Bootcamp I tried to share this passion through hands-on DIY projects with others.

Recently I became more interested in Fine-Tuning the LLM model. I have trained llama2/3, granite on Azure VM. At CS, I want to delve into more advanced AI architecture and innovative computing technologies while solving real-world challenges.

## Cornell University

We all contribute to, and are influenced by, the communities that are meaningful to us. Share how you've been shaped by one of the communities you belong to.

Remember that this essay is about you and your lived experience. Define community in the way that is most meaningful to you. Some examples of community you might choose from are: family, school, shared interest, virtual, local, global, cultural. *(350)

#FTP

A thought-provoking conversation with my little sister made me realise the gap in the Bangladeshi STEM-Education. Driven to make a difference, I founded Future Talent Programme, where I organised a three-month-long program to teach students C++ and Python with hands-on DIY projects with microcontrollers like Arduino and NodeMcu. I also mentored students for national and global hackathons to help them gain practical experience by solving real-world challenges. The team I mentored this year for **NASA-International-Space-Apps-Challenge** made it to the global stage as a Global Nominee.

Little did I know that the bootcamp I started to share knowledge would end up teaching me lifelong lessons and invaluable experiences.

Like my experience with a new batch at FTP'23,  I was teaching like in any other batches:

"….use Wire.beginTransmission() function to send commands to this PCA9685's address."
"Did you understand?" Nabila nodded along with everyone else but her expression betrayed her confusion.

Many students like her in the bootcamp were hesitant to ask questions. I realized my lack of communication and the need to adjust my approach. That's when I introduced the "letterbox" concept, where students could leave anonymous notes on topics to revise in the upcoming class. This method worked as I began receiving a lot of notes.

So far I've taught 150 students each requiring a unique approach to learning and achieved invaluable skills and qualities. For instance, Nabila taught me to be more patient and adapt to different learning paces, while Dipto showed me the importance of positive reinforcement.
Teaching at FTP for the last two years has allowed me to lead with empathy, adapt to challenges, and acquire diverse teaching skills to address every student's needs while recognizing my shortcomings.

At Cornell, I want to take a similar role by volunteering at Cornell's initiatives like the Cornell-Early-STEM-Outreach-Club (GSEC), Cornell-Engineering-High-School-Outreach-Program, and 4-H STEM Programs, where I will have the opportunity to engage with high school and primary kids on STEM activities. Given my diverse experience with kids and high school students from FTP  and navigating the differences and challenges, I am sure I will bring new perspectives to these initiatives.

Fundamentally, engineering is the application of math, science, and technology to solve complex problems. Why do you want to study engineering?*(200)

I started my computing journey with MIT App Inventor at 13. Later I self-taught myself Java in Android Studio to build Android applications. From then on I kept engaging on hackathon and project.

Like my LoRa (SX1278-432 MHz) module-based communication and early disaster prediction system helped me to crack Nasa-Space-Apps which was intended to solve communication issues in Bangladesh during the disaster and provide early prediction.

In my BCI project I created a single channel amplifier with a Raspberry-Pi-Zero to capture neural signals from the visual cortex of the brain. I used my little brother as a test-subject and placed electrodes from the amplifier on the visual-cortex. Processed the signals through **Butterworth filter** algorithm to remove noise, and then used **Fast-Fourier-Transform (FFT)** to analyze the frequency components. Lastly plotted the signal on a line graph for visualization.

I have recently become more interested in fine-tuning LLMs. I have trained Llama 2 and Granite on an Azure VM, which eventually made me a top finalist in an IBM hackathon. In FTP-Bootcamp I tried to share this passion with others.

At CS, I want to delve into more advanced AI architecture and innovative computing technologies while solving real-world challenges.

Why do you think you would love to study at Cornell Engineering?*(200)

While engaging in hackathons and projects over the years, I have realized the importance of collaboration and mentorship. I used to wonder, what if I had been mentored by a professional from this sector? What if I had a companion to collaborate with? For example, in my BCI project, I only recorded neural signals, but BCI technology goes beyond just signal detection.

Cornell's initiatives, like CURB (Cornell-Undergraduate-Research-Board) and CUAI (Cornell-University-Artificial-Intelligence), will provide me with opportunities to collaborate and receive mentor support in research across various fields, including neuroscience, robotics, data science, computer vision, embedded systems, natural language processing, and cyber-physical systems. For someone passionate like me, these opportunities are a playground of endless possibilities.

Most importantly, given my interest and experience in Neuroscience and AI, I want to research NeuroAI at Cornell Neurotech. It’s exciting to see how AI and neuroscience intersect to gain new insights into neural data, generate new hypotheses for understanding the brain and behavior, and create brain-inspired AI architectures.

Additionally, Cornell's 34 student teams, such as Cornell Design-Build-Fly, CU-Autonomous-Drone, and Cornell CUAIR align with my interests in collaboration on different innovative project. It’s like having a platform to turn the ideas into real world solution.

What brings you joy?*

Seeing my project work at the first try, cooking rather than eating; Late-night coding when the surroundings are completely silent; Watching at least one anime a month to recharge myself; Occasionally taking a late night walk when the road is empty for a peaceful moment; Being alone with an orchestral playlist, mesmerizing smell of jesmin at night.

What do you believe you will contribute to the Cornell Engineering community beyond what you've already detailed in your application? What unique voice will you bring? *(100)

“Have you prayed your Salah?” I replied, “No, I don’t pray. I don’t believe in God—I’m an atheist.” In response, people often look at me as though I’m an extraterrestrial being from another planet. Being born in a society where religion is deeply rooted, I have faced countless criticisms like that.

But I have learned to coexist with these differences while genuinely appreciating them.

At Cornell, I aim to take on similar roles by building bridges across differences and creating a community where differences are always appreciated, whether they are related to religion, gender, or any other aspect of identity, no matter how unique or diverse.

## [Duke University]

**What is your sense of Duke as a university and a community, and why do you consider it a good match for you? If there's something in particular about our offerings that attracts you, feel free to share that as well. (250 word limit)***

I started my computing journey with MIT-App-Inventor at 13. Later I self-taught Java to build Android Applications.

My curiosity and passion grew with my age which leads me to explore various fields of CS through projects and hackathons while solving real-world problems.

My LoRa (SX1278-432 MHz) module-based communication and early-disaster-prediction-system helped me to crack Nasa-Space-Apps which was intended to solve communication issues in Bangladesh during the disaster. In my BCI project I created a single-channel amplifier with a NodeMCU to capture neural signals. Recently, I built an autonomous drone with a Raspberry-Pi-Zero navigated by a multimodal model hosted on Azure.

In FTP-Bootcamp I tried to share this passion through hands-on DIY projects with others.

While engaging in hackathon and project over the year I have realized the importance of collaboration and mentorship.

Duke's  Summer Science Undergraduate Research Fellowship (SURF) Program along with MUSER and C-SURF will provide me with opportunities to **collaborate** and get mentor support in research across various fields, including neuroscience, robotics, data science, computer vision, embedded systems, natural language processing, and cyber-physical systems. For someone passionate like me, these opportunities are like a playground of endless possibilities.

But most importantly, given my interest and experience in Neuroscience and AI, I want to research NeuroAI at Duke Institute for Brain Sciences through Summer Neuroscience Program. It's exciting to see how AI and Neuroscience intersect to gain new insights into neural data, generate new hypotheses for understanding the brain and behavior, and create brain-inspired AI architectures.

## Extra Optional

A thought-provoking conversation with my little sister made me realise the gap in the Bangladeshi STEM-Education. Driven to make a difference, I founded Future Talent Programme, where I organised a three-month-long program to teach students C++ and Python with hands-on DIY projects with microcontrollers like Arduino and NodeMcu. I also mentored students for national and global hackathons to help them gain practical experience by solving real-world challenges. The team I mentored this year for NASA-International-Space-Apps-Challenge made it to the global stage as a Global Nominee.

Little did I know that the bootcamp I started to share knowledge would end up teaching me lifelong lessons and invaluable experiences.

I have taught around 150 students over the past two years while encountering countless challenges along the way. It wouldn't be wrong to say I have acquired around 150 skills, as every student requires a unique approach to learning and engagement based on their individual needs and styles.

Teaching at FTP has not only allowed me to lead but has also shown me how to lead with empathy and adaptability while acquiring diverse teaching skills.

At Duke, I want to take a similar role by volunteering at initiatives like the FEMMES, Duke STEM Connect, and other outreach programs, where I will have the opportunity to engage with high school and primary kids on STEM activities. Given my diverse experience with kids and high school students from FTP  and navigating the differences and challenges, I am sure I will bring new perspectives to these initiatives.

## Princeton University

******Important Univeristy For Aid*****

Please describe why you are interested in studying engineering at Princeton. Include any of your experiences in, or exposure to, engineering, and how you think the programs offered at the University suit your particular interests. (Please respond in 250 words or fewer)*

I started my computing journey with MIT-App-Inventor at 13. Later I self-taught Java to build Android Applications.

My curiosity and passion grew with my age which leads me to explore various fields of CS through projects and hackathons while solving real-world problems.

My LoRa (SX1278-432 MHz) module-based communication and early-disaster-prediction-system helped me crack NASA-International-Space-Apps-Challenge which was intended to solve communication issues in Bangladesh during the disaster. In my BCI project I captured neural signals with a single-channel amplifier and NodeMCU. Recently, I built an autonomous drone with a Raspberry-Pi-Zero navigated by a multimodal model hosted on Azure.

In FTP-Bootcamp I tried to share this passion with others through hands-on DIY projects.

While engaging in hackathon and project over the year made me realize that those projects could reach even higher milestones with collaboration and mentorship.

For example, my BCI project is inspired by the concept of Digital-Immortality, which proposes that a person can achieve immortality by mimicking human consciousness in a virtual-world by mapping synaptic network.

To achieve such a feat, I require extensive collaboration and mentorship across multiple disciplines.

This is where programs like StudioLab and Council on Science and Technology (CST) come in, where I can engage with other nerds like me on BCI projects as well as other innovative projects.

But most importantly given my interest and experience on Neuroscience and AI, I want to work in NeuroAI and Intelligent Systems at **Princeton-Neuroscience-Institute**. It is exciting to see how brain-inspired AI can enhance computational architecture and efficiency.

Princeton values community and encourages students, faculty, staff and leadership to engage in respectful conversations that can expand their perspectives and challenge their ideas and be1liefs. As a prospective member of this community, reflect on how your lived experiences will impact the conversations you will have in the classroom, the dining hall or other campus spaces. What lessons have you learned in life thus far? What will your classmates learn from you? In short, how has your lived experience shaped you? (500 words or fewer)*

“Have you prayed your Salah?”. I replied, “No, I don’t pray. I don’t believe in God—I’m an atheist.” In response, people often look at me as though I’m an extraterrestrial being from another planet.

My journey of questioning religious notions began in high school as I delved into books like Sapiens: A Brief History of Humankind by Yuval Noah Harari, The God Delusion by Richard Dawkins, and others. This quote from Sapiens still resonates deeply with me:

"Would the Book of Genesis have declared that Neanderthals descend from Adam and Eve, would Jesus have died for the sins of the Denisovans, and would the Quran have reserved seats in heaven for all righteous humans, whatever their species?"

Even so, in high school, it was mandatory for everyone to take Religion and Moral Studies (Islam), and I did as well, despite being an atheist. I often questioned myself: if I didn't object to it, wasn't that hypocrisy? Yet, at that age, I lacked the courage to openly discuss such a sensitive topic.

I may not be a fan of the tale of Noah's Ark or Adam and Eve from the Quran and Bible but nobody can deny that these stories taught us about brotherhood, truthfulness, how to treat your neighbor, how to care for the elderly, and how to live with compassion and justice—essentially, how a civilized society should behave.

So I have realized there is nothing to be hypocritical about it because these stories, regardless of their origin, still carry valuable lessons that can guide us toward being better individuals.

At the start of this essay, if I were to say that the Quran is one of my favorite books, even as an atheist, you might find it absurd. But now you know.

Apart from that Bengali culture itself is deeply intertwined with religious beliefs and rituals from Islam and other religion, so during my cultural upbringing, I followed many of those practices even without knowing.

I may have my differences with a large portion of my society where religion is deeply rooted but despite these differences, I have come to appreciate the value of diverse beliefs and opinions in my own way. I have learned to coexist with these differences while genuinely appreciating them. I have learned to respect those beliefs, even when I do not share them by engaging respectfully with those who hold different opinions and making an effort to understand the roots of their beliefs.

At Princeton, I aim to take on similar roles by building bridges across differences and creating a community where differences are always appreciated, whether they are related to religion, gender, or any other aspect of identity, no matter how unique or diverse.

Princeton has a longstanding commitment to understanding our responsibility to society through service and civic engagement. How does your own story intersect with these ideals? (250 words or fewer)*

#FTP

"What are you reading?"

"Microcontroller from the ICT book."

"Why are you reading that? Didn't your class teacher demonstrate it in the lab with a microcontroller?"

"They don't even take the ICT class seriously, let alone the lab."

This conversation with my little sister made me realise the significant gap in the Bangladeshi STEM Education. Driven to make a difference in STEM Sector, I founded FTP (Future-Talent-Programme) bootcamp, where I organised an annual three-month-long program to teach students C++ and Python with hands-on DIY projects with microcontrollers like Arduino and NodeMcu. I also mentored students for national and global hackathons to help them gain practical experience by solving real-world challenges. The team I mentored this year for NASA-International-Space-Apps-Challenge made it the global stage as a Global Nominee.

Little did I know that the bootcamp I started to share knowledge would end up teaching me lifelong lessons and invaluable experiences.

I have taught around 150 students over the past two years while encountering countless challenges along the way. It wouldn't be wrong to say I have acquired around 150 skills as every student required a unique approach to learning and engagement based on their individual needs and styles.

Teaching at FTP has not only allowed me to lead but has also shown me how to lead with empathy and adaptability while gaining diverse teaching skills.

At Princeton, I want to apply these skills by volunteering with the Science Outreach Program while engaging in hands-on STEM education with K-12 students.

## Stanford University

**What is the most significant challenge that society faces today?*(max 50)**

Unethical use of AI. The recent AI boom has also influenced social media which is now flooded with AI-generated content like videos and images often used to spread religious or political statements. South-Asia, a hotspot for religious violence, faces a new weapon to spread false narratives and fuel further division.

**How did you spend your last two summers?***

**2023:**

Participated in NSAC (NASA-Space-Apps-Challenge) hackathon; Managed Executive and Technical operations on QbitLab (my tech-startup); created an autonomous drone with Raspberry-Pi-Zero (Fun-Project) ; Freelanced on Fiverr (App & AI);

**2024:**

teaching programming (C++ and Python) with hands-on DIY projects; Participated in the IBM Watsonx hackathon(AI); Competed again on NSAC'24;

**What historical moment or event do you wish you could have witnessed?***

I wish I could have witnessed the moment when Isaac Newton was struck on his head by an apple and thought, “Hey, why did the apple fall straight down?” Imagine his friends nearby, probably like, “Isaac, stop staring at that apple and eat it already!”

**Briefly elaborate on one of your extracurricular activities, a job you hold, or responsibilities you have for your family.***

NASA-International-Space-Apps-Challenge is the world's largest hackathon, where I participated in 2023 and became a global-finalist. I addressed disaster communication challenges with my hardware and software solution. Receiving appreciation for my project on such a platform and knowing it might help countless people someday makes this achievement meaningful to me.

**List five things that are important to you.***

More satisfaction from cooking rather than eating; Late-night coding when the surroundings are completely silent; Watching at least one anime a month to recharge myself; Occasionally taking a late night walk when the road is empty for a peaceful moment; Being alone with an orchestral playlist.

**The Stanford community is deeply curious and driven to learn in and out of the classroom. Reflect on an idea or experience that makes you genuinely excited about learning.* (250)**

I’m Immortal and I will live forever. By immortality, I mean no ageing, an unending life, and witnessing countless history. Sound Absurd? That’s what I even thought when I first encountered Digital Immortality that proposes that if we map all the synaptic connections in the brain and mimic that structure in a virtual-world, we could create a digital representation of a person’s mind, achieving a form of immortality.

This motivated me to work on the Brain-Computer-Interface (BCI) project where I recorded brainwaves or EEGs from the visual cortex of the brain using my own single-channel amplifier. The amplifier could record publication-grade biopotential signals.

I used my little brother as a test-subject and placed electrodes from the amplifier on the visual-cortex by following the International 10-20 system for recording EEG. I utilized a Raspberry-Pi-Zero for real-time data transmission and processed the signals through **Butterworth filter** algorithm to remove noise, and then used **Fast-Fourier-Transform (FFT)** to analyze the frequency components. Lastly plotted the voltage variations over time on a line graph for visualization.

I know this isn't even close to the concept of Digital Immortality but come on it surely counts as a step toward my goal of gaining immortality. With the appearance of NeuroAI, where AI and Neuroscience intersect to gain new insights into neural data, generate new hypotheses for understanding the brain and behavior, and create brain-inspired AI architectures, my vision might come true. I want to engage in research at NeuroAILab in Stanford to accelerate this concept even further.

Virtually all of Stanford's undergraduates live on campus. Write a note to your future roommate that reveals something about you or that will help your roommate – and us – get to know you better.*

Hi there,
This is Sakib. The name Sakib originates from Iran, the Levant, and Bangladesh, and it means "Brightness" or "Star" in Arabic.

I'm a night owl as I have a bad habit of staying up late at night because I find this time very productive for my work due to the quietness.

I occasionally like to take a walk late at night on empty roads because I find it very peaceful. So if I ever ask you to accompany me, don't get worked up. It's just my own way of relaxing.

I have poor vision of -2.5D, so I have to wear glasses, but they feel heavy on my nose, which I find annoying. Most of the time, you'll see me without glasses, but I'll put them on during conversations because everything seems blurry without them.

I like anime a lot. The death of Zero (Lelouch) from *Code Geass* still makes me sad no matter how many times I watch it. So, if you're an anime fan like me, then we'll definitely have plenty to talk about!

I'm very passionate and enthusiastic about technology and innovation, so most of the time, you'll find me working on something or holding a gadget while staring at IDE. I hope we can collaborate on a project someday that might transform the world—after all, even Bill Gates and Paul Allen were roommates when they founded Microsoft.

**Please describe what aspects of your life experiences, interests and character would help you make a distinctive contribution as an undergraduate to Stanford University.* (250)**

#FTP

"What are you reading?"

"Microcontroller from the ICT book."

"Why are you reading that? Didn't your class teacher demonstrate it in the lab with a microcontroller?"

"They don't even take the ICT class seriously, let alone the lab."

This thought-provoking conversation with my little sister made me realise the significant gap in the Bangladeshi STEM Education. Driven to make a difference in STEM Sector, I founded FTP (Future-Talent-Programme) bootcamp where I organised an three-month-long program to teach students C++ and Python with hands-on DIY projects with microcontrollers like Arduino and NodeMcu. I also mentored students for national and global hackathons to help them gain practical experience by solving real-world challenges. This year, the team I mentored for Nasa-Space-App-Chellenge reached the global stage as a Global Nominee.

Little did I know that the bootcamp I started to share knowledge would end up teaching me lifelong lessons and invaluable experiences.

I have taught around 150 students over the past two years while encountering countless challenges. It wouldn't be wrong to say I have acquired around 150 skills, as every student required a unique approach to learning and engagement based on their individual needs and styles.

Teaching at FTP has not only allowed me to lead but has also shown me how to lead with empathy and adaptability while acquiring diverse teaching skills.

At Stanford I want to follow this same path while volunteering at initiatives like **Triple F Empowerment**, and Outreach Program by engaging with K-12 students with STEAM educational activities.

University of Notre Dame

******Important Univeristy For Aid*****

Briefly share what draws you to the area(s) of study you listed.(50)

I started my computing journey with MIT-App-Inventor at 13. My curiosity grew with my age and I began to explore other fields of CS like AI, robotics, IoT, etc., through projects and hackathons.
My major in CS will allow me to delve into more advanced computing technologies and AI architecture.

Everyone has different priorities when considering their higher education options and building their college or university list. Tell us about your "non-negotiable" factor(s) when searching for your future college home.*(150)

My LoRa (SX1278-432 MHz) module-based communication and early-disaster-prediction-system helped me crack NASA-International-Space-Apps-Challenge which solved communication issues in Bangladesh during the disaster. In my BCI project I captured neural signals with a single-channel amplifier and Raspberry-Pi-Zero. Recently, I built an autonomous drone with a Raspberry-Pi-Zero navigated by a multimodal model. Those are few of my projects and hackathons that I engaged over the year.

While engaging on those hackathons and projects made me realize the importance of collaboration and mentorship.

Notre Dame's initiatives like E-SURE, AWARE, NURF, i-SURE, SEEDS, Stinson-Remick Hall of Engineering, Engineering Innovation Hub Program will provide me with opportunities to collaborate and get mentor support in research across various fields, including neuroscience, Robotics, Data Science, computer vision, embedded systems, natural-language-processing, cyber-physical systems with state-of-the-art machinery like Nanofabrication Facility, automated mills, lathes, 3D printers.

For someone passionate like me, these opportunities are like a playground of endless possibilities.

What would you fight for?*(100)

Open any social media app, and I am sure you have witnessed AI-generated content on your feed. The recent AI boom has also influenced social media, which is now flooded with AI-generated content like videos and images, often used to spread religious or political statements. In South Asia, which is a hotspot for religious violence, these AI-generated images are used to spread false narratives and fuel further division by political motives or **religious extremism**. I want to fight against this unethical use of AI, particularly in the religious domain.

Notre Dame's undergraduate experience is characterized by a collective sense of care for every person. How do you foster service to others in your community?*(100)

#FTP

Having a thought-provoking conversation with my sister, who was struggling with her STEM classes from the ICT book, made me realize the significant gap in the Bangladeshi STEM education system. Driven to make a difference in STEM Sector, I founded FTP (Future-Talent-Programme) bootcamp, where I organised an three-month-long program to teach students C++ and Python with hands-on DIY projects with microcontrollers like Arduino and NodeMcu. I also mentored students for national and global hackathons to help them gain practical experience by solving real-world challenges. This year the team I mentored for the Nasa-Space-App-chellenge reached the global stage as a Global-Nominee.

What compliment are you most proud of receiving, and why does it mean so much to you?*(100)

“Damn, Sakib! You said it… and it happened! Amazing boy!” That was one of the few compliments I received after becoming a Global Finalist at the NASA Space Apps Challenge.

Network outages are very common in Bangladesh during disasters, and as a result, crucial aid and medical supplies often can't reach affected areas. I tried to address this issue in the hackathon through my hardware and software solution, SDRAM (Sustainable Disaster Response Alert Mechanism) .

Receiving appreciation for my project on such a platform and knowing it might help countless people someday makes this compliment meaningful to me.

University of Pennsylvania

**Write a short thank-you note to someone you have not yet thanked and would like to acknowledge. (*We encourage you to share this note with that person, if possible, and reflect on the experience!)* (150-200 words)***

**Dear Sifat Bhai,**

I've never had the chance to properly thank you. I still remember that day during my freshman year of high school. I was completely lost—no friends, no companions—sitting alone with a gloomy face. Then you appeared and invited me to join your robotics club. I don't know how you found me, but without your intervention, I wouldn't have had such a great high school experience.

So, thank you for extending your friendship when I was totally alone and lost. Thank you for boosting my confidence and helping me step out of my comfort zone.

I still remember your saying to boost my confidence when I was about to address the club members: *"Whenever you are going to address a big crowd, just assume you are the most knowledgeable person in the room, and the rest don't know anything. This will help you stay calm."*

Guess what? I still follow this advice of yours. From being alone in the class at the start of my high-school and finishing the journey with countless accomplishments and amazing friendships, it happened only because of you. So yes, for the last time, thank-you for being my mentor and guide.

**With love,**
Sakib

**How will you explore community at Penn? Consider how Penn will help shape your perspective, and how your experiences and perspective will help shape Penn. (150-200 words)***

#FTP

A thought-provoking conversation with my little sister made me realise the gap in Bangladeshi STEM Education. Driven to make a difference, I founded FTP (Future-Talent-Programme) bootcamp, where I organised an annual three-month program to teach students C++ and Python with DIY projects with microcontrollers like Arduino and NodeMcu. I also mentored students for national and global hackathons to help them gain practical experience solving real-world challenges. The team I mentored this year for NASA-International-Space-Apps-Challenge made it to the global stage as a Global Nominee.

I have taught around 150 students over the past two years while encountering countless challenges. It wouldn't be wrong to say I have acquired around 150 skills, as every student required a unique approach to learning and engagement based on their individual needs and styles.

At Penn, I want to follow this same path by volunteering at Inveniam Outreach and engaging with K-12 students through STEAM educational activities. Being a part of those diverse community programs will broaden my view and challenge my thinking capabilities. Being a part of the FTP bootcamp taught me how to lead with empathy and adaptability, and I want to bring that to the Penn community.

A thought-provoking conversation with my little sister made me realise the gap in Bangladeshi STEM Education. Driven to make a difference, I founded FTP (Future-Talent-Programme) bootcamp, where I organised an annual three-month program to teach students C++ and Python with DIY projects with microcontrollers like Arduino and NodeMcu. I also mentored students for national and global hackathons to help them gain practical experience solving real-world challenges. The team I mentored this year for NASA-International-Space-Apps-Challenge made it to the global stage as a Global Nominee.

I have taught 150 students over the past two years while encountering challenges along the way. It wouldn't be wrong to say I have acquired 150 skills as every student requires a unique approach to learning and engagement based on their needs and styles.

At Penn, I want to take a similar role by volunteering at Inveniam Outreach programme where I will have the opportunity to engage with high school and primary kids on STEM activities. Given my diverse experience with kids and high school students from FTP and navigating the differences and challenges, I am sure I will bring new perspectives to these initiatives while engaging with these diverse learning environments will broaden my view and challenge my thinking.

Penn Engineering prepares its students to become leaders in technology by combining a strong foundation in the natural sciences and mathematics with depth of study in focused disciplinary majors. Please share how you plan to pursue your engineering interests at Penn. (150-200 words)

To help inform your response, applicants are encouraged to learn more about Penn Engineering and its mission to prepare students for global leadership in technology here. This information will help you develop a stronger understanding of academic pathways within Penn Engineering and how they align with your goals and interests.*

I started my computing journey with MIT-App-Inventor at 13. Later I self-taught Java to build Android Applications.

My curiosity grew with my age and led me to engage in other fields of CS through projects and hackathons.

My LoRa module-based communication and early-disaster-prediction-system helped me to crack Nasa-Space-Apps which was intended to solve communication issues in Bangladesh during the disaster. In BCI project I used a Raspberry-Pi-Zero and single-channel-amplifier to capture neural signals, and recently I built a Raspberry-Pi-Zero powered autonomous drone navigated by a multimodal model. In FTP-Bootcamp I tried to share this passion through hands-on DIY projects with others.

As I engaged in hackathons and projects over the years, I realized that those projects could reach even higher milestones with collaboration and mentorship from multiple disciplines.

This is where **Affiliated Institutes and center like** ASSET,GRASP, PRECISE, PRiML from penn comes in which will provide me oppertunity to enage on research in areas such as Robotics, Data Science, computer vision, embedded systems, natural-language-processing, cyber-physical systems and so on.

For somone nerd like me those oppertunity are like playground of endless possibilities.

LOR Sahed Mam:

December 26, 2024

Dear Undergraduate Admission Committee,

It is my pleasure to write a letter of recommendation for Sakib for your undergraduate program. He was first introduced to me when he came to Blue Bird School & College as a first-year student in 2018, and I was his biology teacher in high school.

As is evident from his transcript, Sakib has excelled throughout his academics with many notable accomplishments which I will not repeat here. I will focus on my experiences with Sakib primarily those related to academics and extracurricular and which demonstrate the qualities of excellent knowledge and leadership.

In addition to considerable intellect and exemplary performances on academics examinations, Sakib is a warm, engaging individual. He was not only hardworking but also consistently active in class and exhibits curiosity and motivation to learn. He comes prepared for all types of learning situations, having researched the relevant topics so that he can participate actively in class. In addition to prioritizing his own learning, he considers the needs of other students by engaging with them on group studies.

He truly excelled in his involvement with the community and volunteer activities. Sakib was deeply connected to various clubs during his high school years. He even participated in several on-campus and off-campus activities. One of the most admirable examples was when he taken programming classes for junior students through Programming Club, showcasing his leadership and dedication.

He also actively engaged on several national competitions, bringing immense pride to our school and further enhancing his reputation as a standout student.

He was praised by teachers for his team management skills as well as his studies. A teacher was so impressed by Sakib knowledge and skills that she invited him to give a speech during our Independence Day ceremony. His speech and presentation was outstanding – comprehensive in scope yet presented efficiently and effectively.

In conclusion, I am happy to give Sakib my highest recommendation for your Undergraduate program and I believe he will be an asset to your undergraduate community.

In my experience, he is in the top 10% of all students with whom I have taught over the past 10 years. If you have any additional questions or require further information, please do not hesitate to contact me.

Sincerely,

Saheda Akter

Assistant Professor, Blue Bird School & College

LOR of Madhab Sir:

I am pleased to write this letter of recommendation for Sakib, who was first introduced to me when he came to Blue Bird School and College as a first year student in 2018. I found Sakib to be a very hard-working, humble, and conscientious student who gets along well with his peers.

He stood out because of his academic achievements and his participation in various co-curricular activities throughout his high school years. As a matter of fact he didn’t limit himself to academics. He actively took part in national competitions, making him stand out from other students in the class. His participation **in** the annual school science fair with different science project showcase his interest in science and technology. He was also involved in club activities with robotics club and was an active member of the science club. I remember him teaching programming to junior students of our school through the programming club. This demonstrates his dedication to both learning and giving back. In conclusion I have known him as a curious, dedicated, and persistent student. His involvement in both academics and extracurriculars highlights his dedication and genuine interest in learning.

It is therefore without reservations that I recommend him to you for any undergraduate program that he may be seeking. I am sure he will be a valuable asset to your organization. Please do not hesitate to contact me if you need further information.

Sincerely yours,
Madhab Ranjan Ray. Lecturer. Physics
Blue Bird School and College

LOR of Shamser Sir:

In 2016-2017, when he was in grades 9-10, I was his math teacher. So, in this context, I have known him for almost 7 years.

In 2016-2017, I was both his math teacher from grades 9-10. He was really kind and easy to get along with. Because of his exceptional teamwork skills, I frequently assigned him as the team leader for numerous group projects and assignments. As a leader, he demonstrated outstanding organizational abilities by successfully delegating responsibilities to his group members, ensuring that everyone contributed to the group's success. His ability to listen to and adopt others' ideas was very impressive. Academically, he regularly displayed a strong grasp of mathematical ideas and addressed problems with curiosity and dedication. He was not afraid to ask questions and sought help when needed, showing a willingness to learn and grow. On a personal level, he was kind and supportive. I have no doubt that he will continue to excel in his future academic endeavors and contribute positively to any community he becomes part of.

I am writing to recommend Sakib Ahmed for your undergraduate program. From 2016 to 2017, I was his math teacher at The Aided High School.

As a student, Sakib always possessed a curious mind and was eager to learn. He was really kind and easy to get along with.

During his time in my class, he engaged in various group assignments where he demonstrated excellent teamwork, leadership skills, and a strong commitment to achieving collective goals.

His ability to listen to and adopt others' ideas was very impressive. Academically, he regularly displayed a strong grasp of mathematical ideas and addressed problems with curiosity and dedication.

On a personal level, he was kind and supportive. I have no doubt that he will continue to excel in his future academic endeavors and contribute positively to any community he becomes part of.

Should you have any further questions about him, feel free to reach me.

Sincerely yours,

Shamser Ali

The Aided High School

Tantipara, Zindabazar, Sylhet, Bangladesh

## Homo Sapiens: The Journey of an Animal That Became God - A Timeline

What if I address you as animal? How will you react!? Probably you will burst into anger, and this is a pretty ordinary response, as referring to someone as an animal often implies they are uncivilized in our society. But try to think for a second. What are we? How do we define animal?

Animals are multicellular, eukaryotic organisms in the biological kingdom Animalia. With few exceptions, animals consume organic material, breathe oxygen, have myocytes, and are able to move, can reproduce sexually, and grow from a hollow sphere of cells, the blastula, during embryonic development.

In this sense, aren't we all animals? Maybe you still have an objection to my statement, as you might say that how can I compare a human being with an animal that has gone through a cognitive revolution about 70,000 years ago? This might be true; we all have the cognitive ability that distinguishes us from other species. But don't the chimpanzees have the same ability? They care for their young in ways comparable to our human society, yet we still classify them as animals. Even we were the same before the cognitive revolution. Maybe a few million years ago we had the same ancestor as the chimpanzees. But after our cognitive revolution we started to see ourselves apart from animals, like an orphan without any siblings or cousins, but most importantly, without any parents.

However what is truly astonishing about this animal is not its cognitive ability but rather how it is exhibiting it. From the innovation of early stone tools to the creation of the modern computer, this creature named Homo sapiens now stands on the verge of becoming a god, prepared to acquire not only eternal youth but also the divine ability of creation and destruction. But there is nothing to be proud of. Maybe we have created big cities, increased food production, created trade networks, and established mighty empires, but we have increased the misery for other animals. In the last few decades we started to become stable and started to make real progress as far as human life is concerned with the reduction of plague, famine, and war. Yet the situation of other animals is deteriorating. Their numbers are decreasing more than ever before. Despite the astonishing things humans are capable of doing, we are still uncertain about our goal. We may have advanced from steamship to space shuttle, but we still don't have any certain goal and don't know where we are approaching. In one word, we are more powerful than ever before, but we have little idea about what to do with this awesome power.

Extra Eassy:

I started my computing journey with MIT App Inventor at 13. Later, I self-taught Java in Android Studio to build Android applications. From then on, I continued engaging in hackathons and projects while pursuing my passion for technology and gadgets.

Network outages are very common in Bangladesh during disasters, and as a result, crucial aid and medical supplies often fail to reach affected areas. I tried to address this issue in the NASA International Space Apps Challenge through my hardware and software solution, SDRAM (Sustainable-Disaster-Response-Alert-Mechanism), where I used a LoRa SX1278 module and NodeMCU connected to my React Native app via a Wi-Fi access point. This system provided communication over 12 km through a half-duplex method. The app also featured a trained sequential neural network to predict lightning and provide alerts for disasters like floods, cyclones, and typhoons. This solution eventually led me to become a Global Finalist in the hackathon.

In my BCI project, I created a single-channel amplifier with a Raspberry Pi Zero to capture neural signals from the visual cortex of the brain. I used my little brother as a test subject and placed electrodes from the amplifier on the visual cortex. I processed the signals through a Butterworth filter algorithm to remove noise, then used Fast Fourier Transform (FFT) to analyze the frequency components. Lastly, I plotted the signal on a line graph for visualization.

I have recently become more interested in fine-tuning LLMs. I trained Llama 2 and Granite on an Azure VM, which eventually made me a top finalist in an IBM hackathon. At FTP-Bootcamp, I tried to share this passion with others.

At LMU, I want to keep pursuing my passion by engaging in research initiatives like the Independent Undergraduate Research Program along with the Summer Undergraduate Research Program (SURP), Research Learning Community, Summer Opportunities for Advanced Research (SOAR), and department-specific research opportunities to collaborate and receive mentorship in research across various fields, including neuroscience, robotics, data science, computer vision, embedded systems, natural language processing, and cyber-physical systems. For someone passionate like me, these opportunities are a playground of endless possibilities.

Most importantly, given my interest and experience in Neuroscience and AI, I want to research the NeuroAI discipline through these initiatives. It’s exciting to see how AI and neuroscience intersect to generate new insights into neural data, formulate new hypotheses for understanding the brain and behavior, and create brain-inspired AI architectures. My passion for technology, programming, and science feels like a magical spell to me—just a few lines of code, and anything is possible. All I need to do is imagine, and with coding, I can bring that imagination into reality. Just like magic. I want to keep pursuing this passion of mine at LMU.

I started my computing journey with MIT App Inventor at 13. Later I self-taught myself Java in Android Studio to build Android applications. From then on I kept engaging on hackathon and project.

Like my LoRa (SX1278-432 MHz) module-based communication and early disaster prediction system helped me to crack Nasa-Space-Apps as Global Finalist which was intended to solve communication issues in Bangladesh during the disaster and provide early prediction.

In my BCI project I created a single channel amplifier with a Raspberry-Pi-Zero to capture neural signals from the visual cortex of the brain. I used my little brother as a test-subject and placed electrodes from the amplifier on the visual-cortex. Processed the signals through Butterworth filter algorithm to remove noise, and then used Fast-Fourier-Transform (FFT) to analyze the frequency components. Lastly plotted the signal on a line graph for visualization.

I have recently become more interested in fine-tuning LLMs. I have trained Llama 2 and Granite on an Azure VM, which eventually made me a top finalist in an IBM hackathon. In FTP-Bootcamp I tried to share this passion with others.

While engaging in hackathon and project over the year I have realized the importance of collaboration and mentorship.

At university, I want to engage in initiatives like Research Experiences for Undergraduates (REU), along with SURF and other department and university-specific research opportunities to collaborate and get mentor support in research across various fields, including neuroscience, robotics, data science, computer vision, embedded systems, natural language processing, and cyber-physical systems. For someone passionate like me, these opportunities are like a playground of endless possibilities.

But most importantly, given my interest and experience in Neuroscience and AI, I want to research NeuroAI at the Neuroscience Laboratory. It's exciting to see how AI and Neuroscience intersect to generate new insights into neural data, formulate new hypotheses for understanding the brain and behavior, and create brain-inspired AI architectures.

Short version:

"Have you prayed your Salah?" I replied, "No, I don't pray. I don't believe in God—I'm an atheist." In response, people often look at me as though I'm an extraterrestrial being from another planet.
Being born in a conservative society where religion and culture are deeply rooted and promoted, I have faced countless criticisms like that.
As I grew up in a bookish family environment, I delved into books like *Sapiens* and started to question those religious notions. This line from the book still resonates with me: "Would the Book of Genesis have declared that Neanderthals descend from Adam and Eve, would Jesus have died for the sins of the Denisovans, and would the Quran have reserved seats in heaven for all righteous humans, whatever their species?"
Bengali culture is deeply intertwined with religious beliefs and rituals from Islam and other religions. So, even as an atheist during my cultural upbringing, I followed many of those practices without even knowing.

I may have differences with a large portion of my society where religion is deeply rooted, but despite these differences, I have come to appreciate the value of diverse beliefs and opinions in my own way. I have learned to coexist with these differences while genuinely appreciating them.
At Smith, I aim to take on similar roles by building bridges across differences and creating a community where differences are always appreciated, whether they are related to religion, gender, or any other aspect of identity, no matter how unique or diverse.

Large version:

"Have you prayed your Salah?" I replied, "No, I don't pray. I don't believe in God—I'm an atheist." In response, people often look at me as though I'm an extraterrestrial being from another planet.

Being born in a conservative society where religion and culture are deeply rooted and promoted, I have faced countless criticisms like that.

As I grew up in a bookish family environment, I delved into books like *Sapiens* and *The God Delusion,* which led me to question many religious notions.

This line from the *Sapiens* still resonates with me:

"Would the Book of Genesis have declared that Neanderthals descend from Adam and Eve, would Jesus have died for the sins of the Denisovans, and would the Quran have reserved seats in heaven for all righteous humans, whatever their species?"

In high school, it was mandatory for everyone to take Religion and Moral Studies (Islam) as a subject and I did as well despite being an atheist. I often questioned myself: if I didn't object to it, wasn't that hypocrisy as I identify myself as an atheist? But at that age, I lacked the courage to openly discuss such a sensitive topic.

I may not be a fan of the tale of Noah's Ark or Adam and Eve from the Quran and Bible but nobody can deny that these stories taught us about brotherhood, truthfulness, how to treat your neighbor, how to care for the elderly, and how to live with compassion and justice and so on. In one sentence how a civilized society should behave.

So I have realized there is nothing to be hypocritical about it because these stories, regardless of their origin, still carry valuable lessons that can guide us toward being better individuals.

At the start of this essay, if I were to say that the Quran is one of my favorite books, even as an atheist, you might find it absurd. But now you know.

Apart from that, Bengali culture itself is deeply intertwined with religious beliefs and rituals from Islam and other religions, so during my cultural upbringing, I followed many of those practices even without knowing.

I may have my differences with a large portion of my society where religion is deeply rooted but despite these differences, I have come to appreciate the value of diverse beliefs and opinions in my own way. I have learned to coexist with these differences while genuinely appreciating them. I have learned to respect those beliefs, even when I do not share them by engaging respectfully with those who hold different opinions and making an effort to understand the roots of their beliefs.

At LMU, I aim to take on similar roles by building bridges across differences and creating a community where differences are always appreciated, whether they are related to religion, gender, or any other aspect of identity, no matter how unique or diverse.

"Have you prayed your Salah?" I replied, "No, I don't pray. I don't believe in God—I'm an atheist." In response, people often look at me as though I'm an extraterrestrial being from another planet.

Being born in a conservative society where religion and culture are deeply rooted, I have faced countless criticisms like this. In high school, I had to take Religion and Moral Studies (Islam) despite not believing in God. Initially, I felt like a hypocrite, questioning why I should study something I didn't follow. However, as I engaged with religious texts and discussions, I realized these stories—Noah's Ark, Adam and Eve—weren't just religious doctrines but also moral lessons on compassion, justice, and humanity.

But I can't take all the credit for this realization. I have to mention Arif, a dear friend of mine with completely opposite beliefs. I still remember our conversations during school breaks about a clause from

Sapiens that contradicts the existence of a god-like entity:

"Would the Book of Genesis have declared that Neanderthals descend from Adam and Eve? Would Jesus have died for the sins of the Denisovans? And would the Quran have reserved seats in heaven for all righteous humans, whatever their species?"

But his response completely caught me off guard: "Bengali culture is deeply intertwined with religious beliefs and rituals from Islam and other religions, which you even followed during your upbringing."

I realized that instead of outright rejecting religious narratives, why not appreciate their ethical teachings and cultural significance?

I may have differences with a large portion of my society where religion is deeply rooted, but despite these differences, I have come to appreciate the value of diverse beliefs and opinions in my own way. I have learned to coexist with these differences while genuinely appreciating them.
At CMC, I aim to take on similar roles by building bridges across differences and creating a community where differences are always appreciated, whether they are related to religion, gender, or any other aspect of identity, no matter how unique or diverse.

NASA Space Apps Challenge is the world's largest hackathon, with 53,000 participants from 185 countries in 2023. I participated in the competition and became a Global Finalist. This achievement even made national headlines in my country: "Horizon BD Represents Bangladesh at NASA International Space Apps Challenge!"

Network outages are very common in Bangladesh during disasters, and as a result, crucial aid and medical supplies often can't reach affected areas. I tried to address this issue in the hackathon through my hardware and software solution, SDRAM (Sustainable-Disaster-Response-Alert-Mechanism) where I have used a LoRa SX1278 module and NodeMCU connected to my React Native app via a Wi-Fi access point. This system provided communication over 12 km through a half-duplex method. The app also featured a trained sequential neural network to predict lightning and provide alerts for disasters like floods, cyclones, and typhoons.

Having my work appreciated on such a global stage, and knowing that it might help countless people someday, makes this achievement especially meaningful to me.

I'm immortal; I won't die. To be specific, I will live forever, like the song says—young forever. By immortality, I mean no aging, an unending life, and witnessing countless moments of history. Sound absurd?

That's what I even thought when I first encountered Digital Immortality that proposes that if we map all the synaptic connections in the brain and mimic that structure in a virtual-world, we could create a digital representation of a person's mind, achieving a form of immortality.

This motivated me to work on the Brain-Computer-Interface (BCI) project where I recorded brainwaves or EEGs from the visual cortex of the brain using my own single-channel amplifier. The amplifier could record publication-grade biopotential signals.

I used my little brother as a test-subject and placed electrodes from the amplifier on the visual-cortex by following the International 10-20 system for recording EEG. I utilized a Raspberry-Pi-Zero for real-time data transmission and processed the signals through Butterworth filter algorithm to remove noise, and then used Fast-Fourier-Transform (FFT) to analyze the frequency components. Lastly plotted the voltage variations over time on a line graph for visualization.

That's it? Yeah, that's it. I know this isn't even close to the concept of Digital Immortality but come on it surely counts as a step toward my goal of gaining immortality. With the appearance of NeuroAI, where AI and neuroscience intersect to gain new insights into neural data, generate new hypotheses for understanding the brain and behavior, and create brain-inspired AI architectures, maybe one day I will make a breakthrough in my search for immortality by researching on NeuroAI and it won't be science fiction anymore as it sounds now.

**My Journey in the Search for Immortality: The Quest Continues**

I'm immortal; I won't die. To be specific, I will live forever, like the song says—young forever. By immortality, I mean no aging, an unending life, and witnessing countless moments of history. Sound absurd?

That's what I even thought when I first encountered Digital Immortality that proposes that if we map all the synaptic connections in the brain and mimic that structure in a virtual-world, we could create a digital representation of a person's mind, achieving a form of immortality.

It was the starting point of a breathtaking journey for me and inspired me so much that I began researching on my own. I encountered a lot of ongoing research, but the closest one was the Blue Brain Project, where researchers aimed to achieve the same but with a mouse brain. Were they successful? Hmm, partially—they managed to model only small portions of the brain due to limited computational resources. So, if we couldn't even successfully model a mouse brain with its 71 million neurons, how can we hope to model a human brain with its 86 billion neurons? I was pretty disheartened and had almost lost all hope. But as the saying goes, a researcher never loses hope. Regardless of the research outcome, I was already deeply motivated.

So what did I do to chase immortality? **The Brain-Computer-Interface (BCI) project where I recorded brainwaves or EEGs from the visual cortex of the brain using my own single-channel amplifier circuit. The amplifier could record publication-grade biopotential signals.**

**I used my little brother as a test-subject and placed electrodes from the amplifier on the visual-cortex by following the International 10-20 system for recording EEG. I utilized a Raspberry-Pi-Zero for real-time data transmission and processed the signals through Butterworth filter algorithm to remove noise, and then used Fast-Fourier-Transform (FFT) to analyze the frequency components. Lastly plotted the voltage variations over time on a line graph for visualization.**

That's it? Yeah, that's it. I know this isn't even close to the concept of Digital Immortality, but come on—it surely counts as a step toward my goal of gaining immortality.

But don't lose hope. Just imagine the era we are living in now, flourishing with technological advancements we never even imagined—like the computer. Who would have thought in the 18th century that we'd innovate something that would revolutionize the world? So, one day might come when immortality becomes a reality, just like computers and cellphones. But I'm a realist in practice, not an idealist who will keep dreaming in the hope that someday someone will come and invent the formula for immortality. No, that's not going to happen with me.

I want to get my hands dirty in the lab and accelerate this concept even further. With the appearance of NeuroAI, where AI and neuroscience intersect to gain new insights into neural data, generate new hypotheses for understanding the brain and behavior, and create brain-inspired AI architectures, maybe one day I will make a breakthrough in my search for immortality and it won't be science fiction anymore as it sounds now. And even you might find yourself living in a virtual world.

I never even imagined discovering a concept so fascinating that I would end up revolving my entire career around it. I want to conclude with this saying of mine: In the medieval age, people searched for immortality in elixirs and mythical quests. But in this age, let's search for immortality in AI and Neuroscience.

## From Sleepless Nights to Global Recognition: My Journey to NASA Space Apps Challenge

*"Congratulations, Sakib! Wishing you a bright future," "Damn Sakib Ahmed! You said it...and it happened! Amazing boy!" were just a few compliments I received after becoming a Global Finalist at the NASA Space Apps Challenge. It is the world's largest hackathon with 58k participants from 185 countries in 2023. This achievement of mine even made the National headline of my Country: "Bangladesh Team Shines Bright at the NASA International Space Apps Challenge! "*

After being selected for the National round in the capital, I nearly had a mini heart attack upon arriving at the venue.

Imagine this: After an 8-hour bus ride to the capital, when I saw the venue filled with tech professionals And there I was a high school graduate, leading a team, it seemed unbelievable. That's what I thought too — there's no chance. I couldn't even imagine competing against those innovative minds. I was trembling with a mix of fear and doubt. But I pulled it off and forced myself to appear confident. Because after all, the rest of the team members were looking at me with expectation, and I needed to make them believe that we could pull it off, no matter how overwhelming it seemed.

For the next 48 hours, it was all about a battle of coding and presentations. As the tech lead, I was responsible for designing the entire system architecture of the application, the DNN model, and the IoT hardware. Additionally, I had to lead the team and ensure progress was synchronized with the presentation team.

After hours of trial and error, debugging, and brainstorming sessions, I started to see my project ideas come to life as working prototypes. Every line of code I wrote and every technical decision I made felt like a step closer to the finish line.

What about Sleep? Not a chance. I barely slept, running on nothing but caffeine. I felt my eyes burning and my brain was exhausted—but I couldn't stop. I had to get it right.

**Network outages are very common in Bangladesh during disasters, and as a result, crucial aid and medical supplies often can't reach affected areas. I tried to address this issue in the hackathon through my hardware and software solution, SDRAM (Sustainable Disaster Response Alert Mechanism). I used a LoRa SX1278 module and NodeMCU connected to my React Native application with a Wi-Fi access point, providing communication through a half-duplex method between 12 km and can be extended by connecting with other LoRa devices in a mesh topology, creating a network.**

**The app also features my trained sequential neural network to predict lightning and provide alerts about disasters like floods, cyclones, and typhoons.**

“Sakib Ahmed from Horizon BD, your team has been recommended for the global nomination!” When I heard this announcement, I was totally dumbfounded.

So, all my hard work, sleepless nights, and doubts had paid off. I had made it to the global stage. But I can’t celebrate just yet—I still have a long way to go.

On November 1st, I received my first email announcing my selection as a global nominee. But the excitement didn’t stop there. On November 29th, I received another email informing me that I was one of the global finalists, selected from 40 teams across the globe.

Horizon BD made it to the Global Finals! This news spread like wildfire across the country, and soon I started receiving messages, calls, and countless mentions on social media filled with congratulations.

When I first started, I never even imagined becoming a Global Nominee, let alone a Global Finalist. I was just a high school boy with his team, trying to appear confident but trembling with fear and doubt about the outcome. But after becoming a Global Finalist, I have realized that it does not matter how professional others may appear. What truly matters is your passion and determination for what you are doing.

**Tell us about your leadership skills in your school or community, and how you plan to continue applying your leadership skills at Stony Brook University.**

In my school and community, I have been fortunate to exhibit strong leadership skills that have had a positive impact on those around me and  my surroundings. Leadership is not just about holding a high-ranking position or making all the important decisions. It is about being able to motivate, inspire, and guide others in a way that helps everyone achieve their best. I believe I achieved these important leadership qualities and plan to continue applying them at the **Stony Brook University**.

A pivotal experience that shaped my leadership skills was founding the Future Talent Programme (FTP) bootcamp, where I organized an annual three-month-long program to teach students C++ and Python through hands-on DIY projects with microcontrollers like Arduino and NodeMCU to address the significant gap in Bangladeshi STEM education. I also mentored students for national and global hackathons to help them gain practical experience by solving real-world challenges. The team I mentored this year for the NASA International Space Apps Challenge reached the global stage as a Global Nominee.

Little did I know that the bootcamp I started to share knowledge would end up teaching me lifelong lessons and invaluable experiences.

I have taught around 150 students over the past two years while encountering countless challenges along the way. It wouldn’t be wrong to say I have acquired around 150 skills, as every student requires a unique approach to learning and engagement based on their individual needs and styles. For instance, Nabila taught me to be more patient and adapt my teaching methods to fit different learning paces, while Dipto taught me the importance of positive reinforcement. Arif inspired me to explore creative ways of explaining complex concepts, such as using analogies or visuals to make learning more engaging, to name a few.

Apart from my engagement in the community, I actively participated in hackathons and competitions to solve real-world challenges while leading my team in prestigious events like the NASA International Space Apps Challenge, Microsoft Imagine Cup, and IBM TechExchange Hackathon, to name a few. In the NASA Space Apps Challenge, I addressed communication issues in Bangladesh during disasters.

Network outages are very common in Bangladesh during disasters, and as a result, crucial aid and medical supplies often fail to reach affected areas. I tried to address this issue in the hackathon through my hardware and software solution, SDRAM (Sustainable Disaster Response Alert Mechanism), which eventually led me to become a global finalist in that hackathon.

Additionally, as I discuss my leadership roles, I cannot forget about QbitLab, a startup I co-founded and currently lead as its CEO. Founded in 2023, QbitLab is a tech startup focused on technology, information, and education. Under my leadership, my team delivered solutions via Fiverr and developed SaaS products in sectors such as healthcare, gaming, environmental sustainability, education, and XR, generating $20K in revenue. These products have an active user base of 15,000 customers. FTP bootcamp itself is a project founded by QbitLab, where I lead my team from QbitLab to mentor students alongside me.

Looking back, engaging with these programs, hackathons, and institutions while leading multiple teams in synchronization has significantly shaped my leadership journey. It taught me how to effectively manage diverse groups, delegate responsibilities, and ensure seamless collaboration to achieve common goals. These experiences refined my organizational skills, enhanced my ability to communicate clearly.

For example Teaching at FTP for the last two years has allowed me to lead with empathy, adapt to challenges, and acquire diverse teaching skills to address every student's needs while recognizing my shortcomings. By participating in hackathons and solving real-world challenges through technology, I honed my ability to innovate under pressure, think critically, and collaborate effectively with diverse teams. Most importantly, I improved my technological expertise. Similarly, in QbitLab, leading a startup taught me strategic decision-making, team management, and the importance of aligning technology solutions with real-world problems to create impactful outcomes.

For me, those skills were as diverse as the experiences themselves, with each one stemming from different sectors such as community engagement, business, or technology. At Stony Brook, I hope to put this diverse skill set to work by engaging in initiatives like the Center for Service Learning and Community Service, as well as other collaborative opportunities, to address the needs of surrounding communities through STEM initiatives or by creating a powerful impact similar to FTP by teaching and mentoring.

## 1. Presidential Scholars represent the essence of Clark. What about Clark University- programs, community, culture, you name it- and about your own personality or values prompted you to consider becoming a Clarkie? In short: why Clark?

I started my computing journey with MIT App Inventor at 13. Later, I self-taught Java in Android Studio to build Android applications. From then on, I continued engaging in hackathons and projects while pursuing my passion for technology and gadgets.

Network outages are very common in Bangladesh during disasters, and as a result, crucial aid and medical supplies often fail to reach affected areas. I tried to address this issue in the NASA International Space Apps Challenge through my hardware and software solution, SDRAM, which eventually led me to become a Global-Finalist in the hackathon. The research paper I wrote on this system as a new approach to addressing lightning, titled "A Deep Neural Network Approach for Highly Efficient Lightning Prediction and Warning Systems," has been accepted for presentation at the prestigious 8th International Conference on Engineering Research, Innovation, and Education (ICERIE 2025).

The recent boom in the AI sector piqued my interest as well, and I immersed myself in it out of curiosity. This led me to fine-tune models like Llama 2, Llama 3, and Granite on an Azure VM, which eventually helped me become a top finalist in an IBM hackathon.

At the FTP Bootcamp, I shared this passion of mine with others. It was a three-month-long program to teach underserved students C++ and Python through hands-on DIY projects using microcontrollers like Arduino and NodeMCU. I also mentored students for national and global hackathons to help them gain practical experience by solving real-world challenges. This year, the team I mentored for the NASA Space Apps Challenge reached the global stage as a Global Nominee.

The fact that Clark University is a private research university means that I have the opportunity to become more than a mere student. The university allows me to cross-enroll in courses at other institutions within the Higher Education Consortium of Central Massachusetts (HECCMA), enabling me to keep pursuing this passion by engaging in research initiatives to collaborate and receive mentorship in research across various fields, including neuroscience, robotics, data science, computer vision, embedded systems, natural language processing, and cyber-physical systems. For someone passionate like me, these opportunities are a playground of endless possibilities.

Additionally, the 130+ student clubs and organizations at Clark will provide me with the opportunity to not only explore my passion for science and technology but also share it through clubs like ASCEND in STEM. Clubs such as the ACM Student Chapter and Clark Competitive Computing Club (C4) accelerate these opportunities even further by providing platforms to engage in computer science-related activities, while Clark's Center for Technology, Innovation, and Entrepreneurship will allow me to explore the intersection of technology and entrepreneurship. That’s why I strongly believe Clark is the right environment for me to foster my passion for technological innovation and collaboration, and ultimately make a meaningful impact in my community by addressing global challenges like food security, education, and healthcare through innovation, just as I have done in disaster communication and smart management systems through SDRAM.

## 2. Describe a learning experience that was particularly meaningful to you. Why does this moment of learning stand out to you? What did you learn about yourself as a learner or about your academic journey and future aspirations through this learning experience?

“Congratulations, Sakib! Wishing you a bright future,” “Damn, Sakib Ahmed! You said it...and it happened! Amazing boy!”—these were just a few of the compliments I received after becoming a Global Finalist at the NASA International Space Apps Challenge. It is the world's largest hackathon, with 58,000 participants from 185 countries in 2023. This achievement even made the national headlines in my country: “Bangladesh Team Shines Bright at the NASA International Space Apps Challenge!”

Network outages are very common in Bangladesh during disasters, and as a result, crucial aid and medical supplies often can't reach affected areas. I addressed this issue during the hackathon through my hardware and software solution, SDRAM, where I used a LoRa SX1278 module and NodeMCU connected to my React Native app via a Wi-Fi access point. This system provided communication over 12 km using a half-duplex method. The app also featured a trained sequential neural network to predict lightning and provide alerts for disasters like floods, cyclones, and typhoons.

Having my work appreciated on such a global stage, and knowing that it might save countless lives someday, made this achievement especially meaningful to me.

But the path to reaching that milestone was challenging, as I had to navigate through many competitive stages. As a result, I gained a deeper understanding of problem-solving, developed a heightened sense of strategic thinking, and honed my team management skills.

The local event in Dhaka, which lasted for 48 hours, was a battle of coding and presentations. As the tech lead, I was responsible for designing the entire system architecture of the solution, from the DNN model to the IoT hardware. Additionally, I had to lead the team and ensure that progress was synchronized with the presentation team. After hours of trial and error, debugging, and brainstorming sessions, I began to see my project ideas come to life as working prototypes. However, I couldn't bring myself to take a nap, as every second counted. Instead, I kept myself awake with a few doses of caffeine.

When I first started, I never even imagined becoming a Global Nominee, let alone a Global Finalist. As I honed my team management skills, I came to realize that a team is like a complex system of interconnected gears, both big and small. As long as it is well-coordinated, it can achieve anything, no matter how impossible it seems at first. The trial-and-error process during the hackathon also helped me foster my problem-solving skills on a new level. While working under pressure, I learned to remain calm, think critically, and make quick, informed decisions, ensuring steady progress even in the face of unforeseen challenges.

Looking back, this hackathon played a important role to shape how I approach challenges, collaborate with others. It not only boosted my confidence but also inspired me to pursue future hackathons and initiatives, equipping me with skills like team management, strategic planning, and problem-solving, which have made future collaborative tasks much easier.

## How will you contribute to the Pinnacle/Clark Scholars program, and how does it align with your future goals?

I started my computing journey with MIT App Inventor at 13. Later, I self-taught Java in Android Studio to build Android applications. From then on, I continued engaging in hackathons and projects while pursuing my passion for technology and gadgets.

One significant project was the development of the Sustainable-Disaster-Response-Alert-Mechanism (SDRAM). Network outages are very common in Bangladesh during disasters, and as a result, crucial aid and medical supplies often fail to reach affected areas. I tried to address this issue in the NASA-International-Space-Apps-Challenge through my hardware and software solution, SDRAM, where I used a LoRa SX1278 module and NodeMCU connected to my React Native app via a Wi-Fi access point providing communication over 12 km through a half-duplex method while also featuring a sequential neural network to predict lightning and provide alerts for disasters like floods, cyclones, and typhoons. This solution eventually led me to become a Global Finalist in the hackathon.

But I wanted to share this passion of mine with others, which is when I founded the Future-Talent-Programme (FTP) bootcamp, where I organized an annual three-month-long program to teach students C++ and Python through hands-on DIY projects with microcontrollers like Arduino and NodeMCU while also addressing the significant gap in Bangladeshi STEM education. The team I mentored this year for the NASA-International-Space-Apps-Challenge reached the global stage as a Global Nominee.

Through the Pinnacle/Clark Scholars Program, I want to keep pursuing my passion by engaging in research initiatives while making a meaningful impact in the community, like FTP. I also aim to contribute to the program by collaborating with faculty members on innovative projects that address global challenges like food security, education, and healthcare through innovation, just as I have done in disaster responce and smart management systems through SDRAM.